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Studying the micromechanical behaviors of a polycrystalline metal by artificial neural networks

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ABSTRACT

Though physics-based crystal plasticity models are able to accurately predict the material response under complex loading conditions, the high computational cost limits their engineering applications. Machine learning techniques, widely used to understand and predict data trends, can provide advantageous computational efficiency over conventional numerical techniques. In the current work, an artificial neural network (ANN) model is proposed and trained based on the datasets generated by the physics-based viscoplastic self-consistent (VPSC) model on the material behavior of OFHC copper. The ANN model successfully captures the loading path-dependent micromechanical behavior of the copper polycrystals with arbitrary texture, even beyond the bounds of the generated dataset. The proposed ANN model, without any major loss in accuracy, can greatly improve the computational efficiency. It can be envisioned that the engineering application of the physics-based model becomes promising through alternatively using the ANN model, which treats the underlying complex physics as a blackbox.

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1. Introduction

The mechanism-based analysis of plastic deformation is a typical scientific research method used for understanding the system and systematically established relationships. Many disciplines are based on analytical, statistical, and numerical mechanism-based models to interpret results, understand fundamental relationships, and predict the unknowns. The occurrence of a physical phenomenon is usually complicated and governed by multiple simultaneously acting mechanisms. However, the laws used and the resulting mechanism-based models are sometimes idealized and simplified [1-4].

Instead, data-driven models, which attract extensive attention in recent years, are not necessarily aware of the fundamental mechanisms but require a large amount of data as knowledge to learn the complicated relationships. Even a very complicated task can be accomplished as long as the model is trained enough to master the rules behind. Moreover, this model saves a lot of computational resources compared to the mechanism-based model, since the required computational resource for prediction is small once training has been accomplished. Most of the data-driven models are based on machine learning (ML) and data mining. ML techniques have been applied to different fields such as flexible manufacturing systems, biology, and medicine [5-7].

Understanding the mechanical behavior of metallic materials is important in the advancement of manufacturing technology. It is well acknowledged that the microstructure decisively determines the behavior of a material. As one mechanism-based model for a polycrystalline material, the crystal plasticity model bridges the behaviors to the microstructure, in terms of deformation slip, deformation twinning, grain orientation, grain-grain interaction, etc. Many polycrystal plasticity models have been developed and applied to various polycrystalline materials and significant insights have been gained for different materials [8-14]. The crystal plasticity model, rooted in theory-based hypothesis formation and experiment-based testing, is inherently a trial-and-error approach and usually time-consuming. Finding a suitable model and refining it until it produces the desired results is often a lengthy process. Moreover, the use of these models in engineering applications is limited due to their high computational cost. Therefore, we consider using ML to understand the mechanical behavior of materials in the present work. ML requires a large dataset that is not amenable to the currently existing experimental data. Hence,

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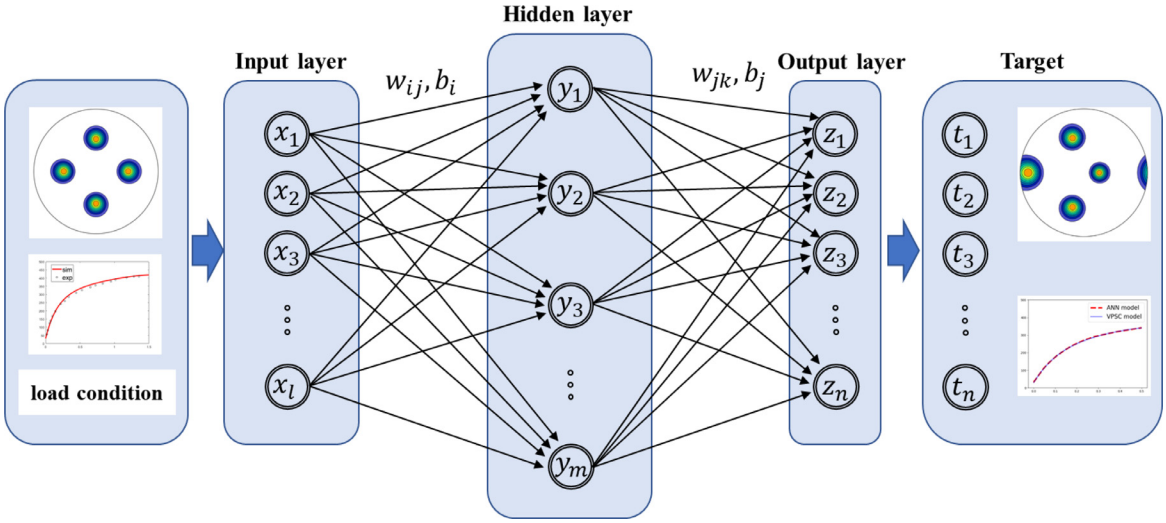


Fig. 1. Schematic of the proposed ANN framework used in the present work.

this requirement is overcome by using datasets based on crystal plasticity simulations. Such an approach is expected to retain the physics-based constitutive response, while considerably speeding up the computational simulations.

With the advent of material informatics, ML has been used to study the impact of microstructural features on stress hotspot for the face-centered cubic (FCC) and hexagonal close-packed (HCP) materials [15,16]. The decision tree-based model was utilized to explore the driving force behind deformation twinning in magnesium alloy AZ31 [17]. ML techniques were applied to establish the relationship between material properties and material compositions along with processing conditions. The obtained predictive model was used to achieve minimum loss and maximum magnetic saturation of Fe-based soft magnetic materials [18]. Deep learning-based models were proposed to study process-structure-property linkages in complex materials systems from a variety of materials data [19]. The artificial neural network (ANN) model was used to build a relation between the composition and processing variables with mechanical properties, and the inverse design of target age-hardenable properties was realized [20]. On the other hand, several attempts have been made to study the microstructure and mechanical response by using ANN models. In these works, ANN models, such as convolutional neural network (CNN) and graphic neural network (GNN) were used to extract low-dimensional descriptors from microstructures, where the grains interact strongly. These descriptors can enhance the supervised learning algorithm to achieve accurate response prediction. The evolution of stress field can be predicted by using a neural network with convolutional layers encoding correlations in time and space [21]. A hybrid convolutional-recurrent neural network was utilized to predict mechanical properties of polycrystalline materials by correlating the underlying crystalline structure with the stress response [22]. A machine learning approach that integrates geometric deep learning and Sobolev training was proposed to generate a family of finite strain anisotropic hyperelastic models that predict the homogenized responses of polycrystals previously unseen during the training [23]. The ANN model was also employed to capture the polycrystalline material behavior and microstructure evolution based on the data generated by crystal plasticity finite element simulations [24]. However, in the latter work, the maximum strain was limited to relatively small ($\leq$30%), and therefore the texture evolution was not drastic. In addition, because the training, validating, and testing datasets were based on a prescribed initial texture, its applicability to a material with an arbitrary texture was not established. However, this extension is deemed important for the wide application of a data-driven model in industrial manufacturing.

In the present work, an ANN-based model is proposed to predict the polycrystalline material behavior and microstructure evolution of a face-centered cubic (FCC) polycrystalline material OFHC copper. The strain behavior of OFHC copper under different load conditions has been measured and simulated by crystal plasticity models [25-27]. In this work, the ANN-based ML framework is used in conjunction with validated crystal plasticity data to predict the stress-strain behavior and texture evolution in OFHC copper under uniaxial tension, uniaxial compression, and simple shear. To obtain reliable datasets for the proposed ANN model, VPSC simulations were performed and validated with experimental results. Next, stress-strain, load conditions, and texture data from VPSC simulations of OFHC copper were used to train, validate, and test the ANN model. The ANN model showed excellent agreement with VPSC simulations under different load conditions. The ANN model was subsequently used to perform predictions outside the boundaries of the original dataset. The results show that the trained ANN model can capture the mechanical behavior and texture evolution of materials well. Moreover, the runtime comparison between ANN and VPSC simulations showed important computational savings, and it highlights the computational advantage of using ANN models over conventional numerical simulations.

2. Problem formulation

2.1. Artificial neural network (ANN) model

Artificial neural networks (ANNs) can mimic nonlinear, complicated, and implicit relationships by applying numerous nonlinear processing units (i.e., neurons). Based on adequate training, the learned relationship from incomplete and disorderly information can be used to generalize rules to new cases. The structure of an ANN is hierarchical with neurons categorized into an input layer, multiple hidden layers, and an output layer (Fig. 1), where the network evolves to improve the predictability of the massively interconnected neurons. Processed data can be obtained from the ANN structures through hidden and output layers based on inputs entered by the input layer. In the mathematical theory of ANNs, the universal approximation theorem states that a feed-forward network with a single hidden layer containing a finite number of neurons can approximate continuous functions under mild assumptions on the activation function [28]. Usually, a maximum of three

hidden layers is sufficient to solve most of the problems. ANNs were extensively used for solving various material science problems, such as predicting material properties, optimizing processing parameters, etc. [20,29].

In this work, we establish an ANN model with a feedforward multi-layers perceptron type architecture trained by the back-propagation (BP) algorithm [30]. The activation function of the neurons between layers is selected as a hyperbolic tangent function [31], which leads to the following relation between the first hidden layer and the input layer:

$$y_i = \frac{2}{\left(1 + e^{-2(w_{ij}x_j + b_i)}\right)} - 1 \tag{1}$$

where w_{ij} and b_j are the weight and bias associated with x_j and y_i, respectively. The same relation as Eq. (1) is applied to other adjacent layers but with their own weights and biases until the output vector z is obtained. The more the layers are applied, the more nonlinear the relationship between the outputs and inputs will be. The relationship between the output vector z and the input vector x in an ANN model can be assumed by an implicit function F, i.e., $z = F(x)$. The function F is found by minimizing the loss function that relates to the difference between the output vector z and the target value t. In the present work, the mean square error (MSE) is used for the loss function:

$$E_M = \frac{1}{n}\sum_{i=1}^{n}(t_i - z_i)^2 \tag{2}$$

2.2. ViscoPlastic Self-Consistent (VPSC) model

The applicability and the quality of any ML prediction technique rely significantly on having a plentiful and accurate dataset. Therefore, sample data must be carefully collected for the designed ANN model. Also, the dataset should contain sufficient information that is representative and descriptive of the mechanical behavior of the material. To obtain representative and reliable datasets for the ANN model, simulations based on the visco-plastic self-consistent (VPSC) polycrystal plasticity model [32] were performed and validated with experimental results of OHFC copper. Datasets regarding the stress, strain, and grain orientations (texture) for the ANN model are generated based on the VPSC simulations with the validated parameters.

The VPSC model is described briefly here for completeness. The model treats the polycrystalline aggregate as a homogeneous effective medium (HEM), and each grain as an ellipsoidal inclusion embedded in the HEM. The correlation between the inclusion and the HEM can be obtained through the classical Eshelby inclusion formulism. Consequently, though the exact spatial correlation between grains is not accounted for, the interaction between grains is indirectly accounted for through the HEM. The VPSC model neglects the elastic deformation because its contribution is trivial at large deformation. The deformation due to the operation of the 12 $\{111\}\langle 1\bar{1}0\rangle$ slip systems can be expressed as

$$\dot{\varepsilon}_{ij} = \sum_{\alpha}\dot{\gamma}^{\alpha}S_{ij}^{\alpha} \tag{3}$$

where S_{ij}^{α} is the Schmid tensor composed by the slip direction (s_i^{α}) and slip plane normal (n_i^{α}) as $S_{ij}^{\alpha} = (s_i^{\alpha}n_j^{\alpha} + s_j^{\alpha}n_i^{\alpha})/2$, and $\dot{\gamma}^{\alpha}$ is the shear rate for the slip system α:

$$\dot{\gamma}^{\alpha} = \dot{\gamma}_0\left|\frac{\tau^{\alpha}}{g^{\alpha}}\right|^{\frac{1}{m}} sgn(\tau^{\alpha}) \tag{4}$$

where $\dot{\gamma}_0$ is a reference shear rate, m is the strain-rate sensitivity, τ^{α} and g^{α} are the resolved shear stress (RSS) and critical resolved shear stress (CRSS) of slip system α, respectively. The RSS is computed from the Cauchy stress by $\tau^{\alpha} = \sigma_{ij}S_{ij}^{\alpha}$. g^{α} evolves with the accumulated shear strain $\Gamma = \sum_{\alpha}\int|\dot{\gamma}^{\alpha}|dt$ by:

$$\dot{g}^{\alpha} = \frac{d\hat{g}^{\alpha}}{d\Gamma}\sum_{\beta}\theta^{\alpha\beta}\left|\dot{\gamma}^{\beta}\right| \tag{5}$$

where $\theta^{\alpha\beta}$ are the coupling coefficients to account for the latent hardening between systems α and β. $\hat{g}^{\alpha}$ is defined by an extended Voce law:

$$\hat{g}^{\alpha} = g_0^{\alpha} + (g_1^{\alpha} + \theta_1^{\alpha}\Gamma)\left(1 - \exp\left(-\frac{\theta_0^{\alpha}}{g_1^{\alpha}}\Gamma\right)\right) \tag{6}$$

where g_0^{α} and $g_0^{\alpha} + g_1^{\alpha}$ are the initial and back-extrapolated threshold CRSSes, θ_0^{α}, and θ_1^{α} are the initial and asymptotic hardening rates, respectively. According to the previous study, $\theta^{\alpha\beta} = 1$ can obtain good results [25] and is employed in the present work. The grain orientation is represented by the rotation tensor $R_{ij}(t)$ in the model, which is updated incrementally by the lattice spin ω_{ij} [33],

$$R_{ij}(t + \Delta t) = (\delta_{ik} + \omega_{ik}\Delta t)R_{kj}(t) \tag{7}$$

Table 1
Voce Hardening law parameters for OFHC copper

Mode	τ_0(MPa)	τ_1(MPa)	θ_0(MPa)	θ_1(MPa)
$\{111\}\langle 1\bar{1}0\rangle$	16.0	162.0	360	5

2.3. Parameter calibration for VPSC model

The material studied in the present work is the OFHC copper with the most complete experimental data [26]. Before generating the datasets for the ANN model, the hardening parameters for the OFHC copper are calibrated against the experimental data. The initial texture of the OFHC copper is represented by a set of 1000 grains with random orientations (see inset the {111} pole figure in Fig. 2). The reference shear rate $\dot{\gamma}_0$, and the rate sensitivity, m, prescribed the same for all slip systems, are taken as $0.001s^{-1}$ and 0.012, respectively [34]. The experimental stress-strain curve under uniaxial tension is reproduced by the model (run under the affine interaction) (Fig. 2) The best-fitted hardening parameters are listed in Table 1 and are subsequently used to generate the datasets for the proposed ANN model.

2.4. Datasets generation

The dataset for the proposed ANN model includes the texture (represented by 96 grains), loading condition, strain, and stress. Inputs to the ANN model comprises the strain, loading condition, and the initial texture represented by sets of three Euler angles $\{\varphi_1, \phi, \varphi_2\}$ and their weights. Three typical loading conditions, uniaxial tension, uniaxial compression, and simple shear are considered in the present study. The associated velocity gradients are listed in Table 2.

In order to properly predict the mechanical response of OFHC copper with an arbitrary initial texture via machine learning, several possible textures must be sampled. This poses a difficulty, as an individual grain can orient randomly in the Eulerian space, which brings about massive combinations of the grain orientations for the OFHC polycrystal. In the present study, sampling textures were generated with 96 orientations randomly distributed in the Eulerian space and with their weights varying. Through varying the weights, textures with a single-crystal (with only one non-zero weight), bicrystal, strongly textured, and weakly textured OFHC coppers are obtained. A total of 1221 initial textures are created

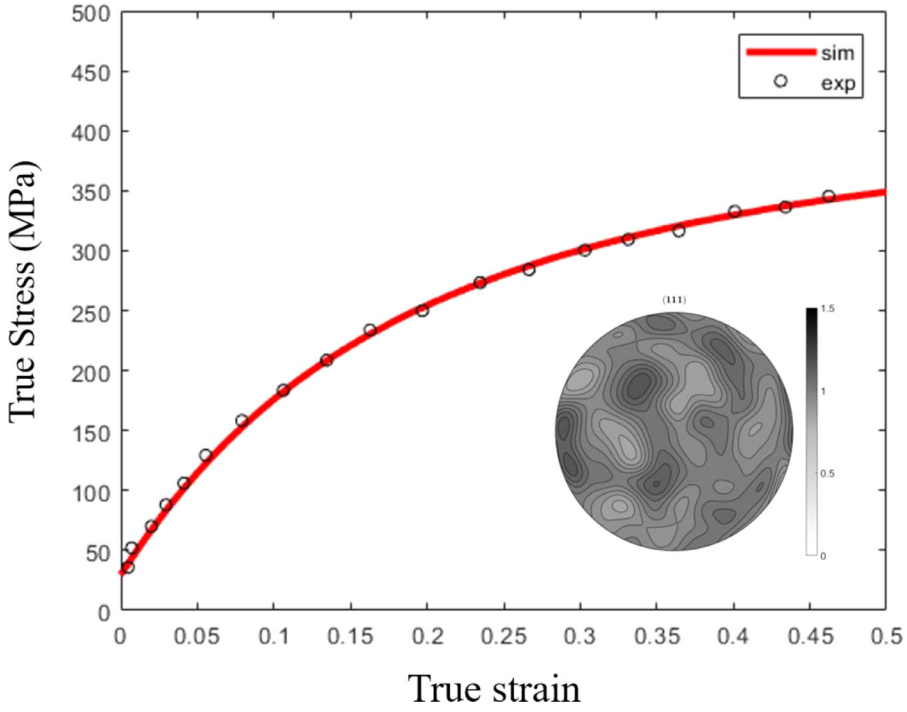


Fig. 2. Experimental and simulated stress-strain curves under uniaxial tension. The inset is the initial texture of the OFHC copper in terms of {111} pole figure.

Table 2
The boundary conditions of the three typical loading conditions

Tension	Compression	Simple shear
$\dot{\varepsilon}_{11} = 0.001/s$; $\sigma_{ij} = 0$ except for σ_{11}	$\dot{\varepsilon}_{11} = -0.001/s$; $\sigma_{ij} = 0$ except for σ_{11}	$\dot{\varepsilon}_{ij} = 0$ except for $\dot{\varepsilon}_{12} = \dot{\varepsilon}_{21} = 0.001/s$

and used for VPSC calculations under three loading conditions. Totally, there are $3 \times 1221 \times$ *number of increments* sample data generated for the ANN model.

However, the number of grains also brings over-fitting problems to the ANN model to a certain extent. Therefore, the dimensionality reduction technique, principal component analysis (PCA) [35,36], is applied to capture the characteristic texture information. PCA is a statistical technique for simplifying the complexity of high-dimensional data while maintaining its variance to retain more information about data. The principal components obtained are orthogonal, which means they are uncorrelated. Furthermore, they are ranked in order of their explained variance or the information they contain. The first principal component (PC1) contains the most information in the dataset, PC2 contains the second-most information, and so on. Therefore, the dimensionality can be reduced by selecting an appropriate number of principal components based on the cumulative explained variance. Therefore, the dimension of each texture can be reduced by using the principal components which capture the main characteristic of texture. Fig. 3 shows the PCA reduction based on thousands of textures which includes the initial textures and the developed textures during deformation. Apparently, the explained variance ratio increases with the number of principal components to reconstruct textures. As shown in Fig. 3, 200 principal components are sufficient to completely represent the raw texture. Therefore, the dimension of each texture ($4 \times 96 = 384$) is reduced to the number of principal components (200) in the present work.

It is important to mention that the three loading conditions listed in Table 1 are encoded with one-hot and introduced into the ANN model. Therefore, such a well-trained ANN model is capable of predicting the texture evolution and mechanical response under all three loading conditions without retraining. As illustrated in Fig. 1, the inputs of the ANN model include the strain history, the initial texture information after PCA dimensionality reduction, and the encoded loading condition. The outputs of the ANN model include the stress and the deformed texture. Two hidden layers with 512 neurons are employed in the ANN model, which results in a total of 474624 trainable parameters.

It is common for larger networks to more easily overfit the training data. In this work, classic regularization techniques in neural networks are utilized. Leveraging some basic approaches like dropout [37,38], norm regularization, and early stopping can dramatically improve performance with minimal effort. Dropout, a simple approach to avoid overfitting by randomly skipping some nodes in a net during each training iteration. In this work, the dropout rate is select 0.2. Early stopping strategy monitors the performance of the model for every epoch on the validation set during the training and terminates the training conditional on the validation performance. In this work, the early stopping strategy halts training when the model's validation error hasn't measurably improved for 8 epochs. L1 regularization adds a penalty on the different parameters of the model to reduce the freedom of the model. Hence, the model will be less likely to fit the noise of the training data and will improve the generalization abilities of the model.

3. Results and discussion

In the process of generating the datasets, the primary strain is imposed to 0.5 in 100 increments. Therefore, $1221 \times 3 \times 101 = 369963$ data samples are generated for the establishment of the ANN model. These datasets were parsed randomly with 70% training set, 15% validating set, and 15% testing set

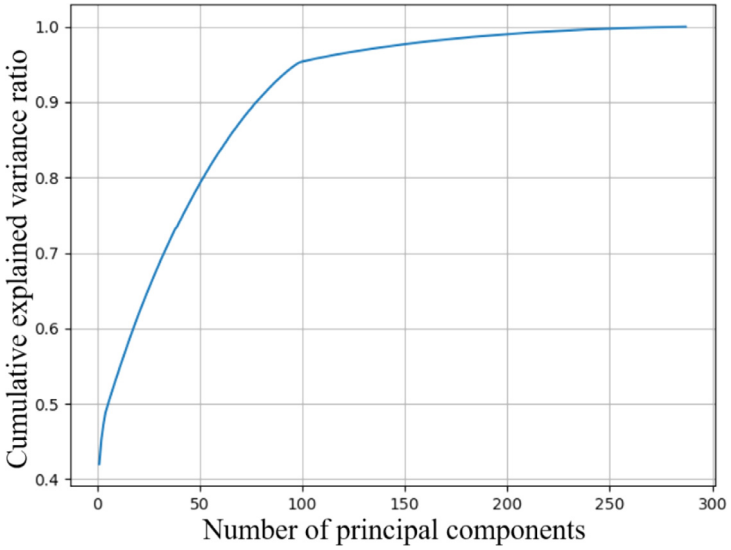


Fig. 3. The relationship between the explained variance ratio and the number of principal components

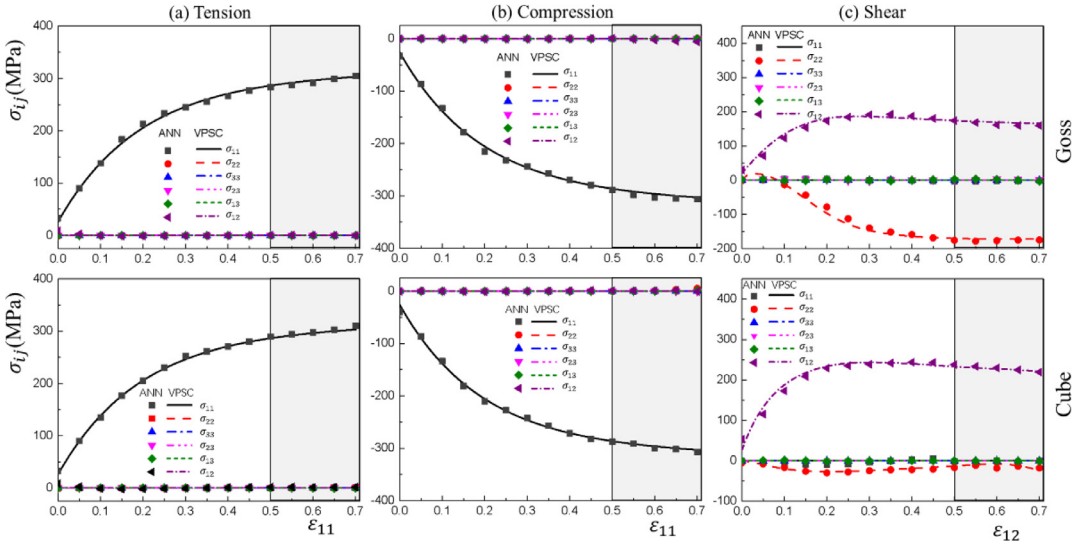


Fig. 4. Comparison between VPSC and ANN stress-strain curves under (a) uniaxial tension, (b) uniaxial compression, and (c) simple shear for Goss (top row) and Cube (bottom row) single crystals.

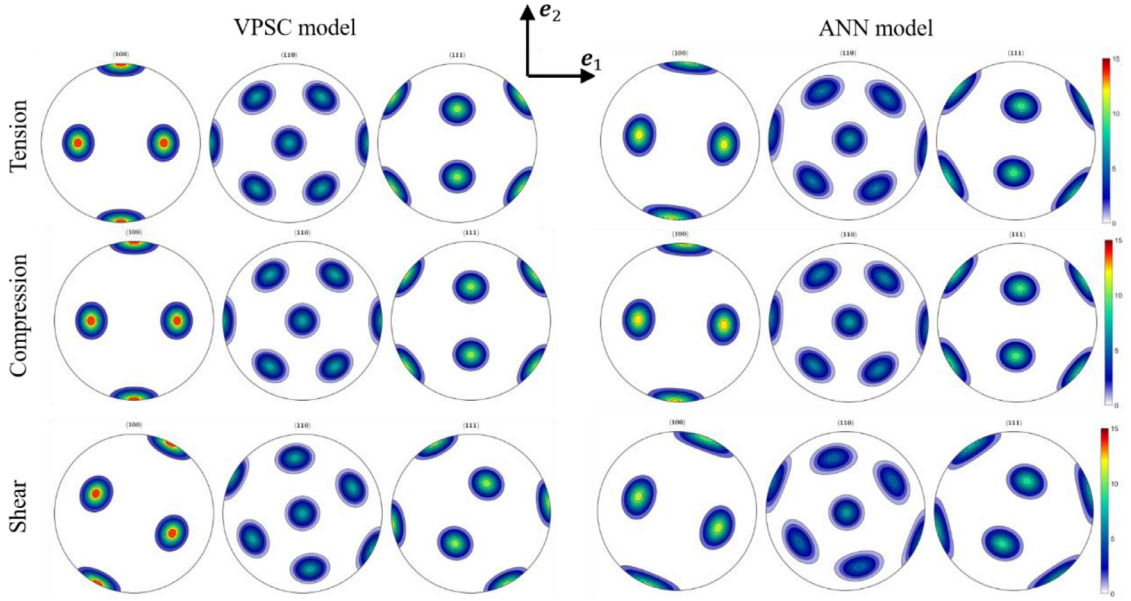


Fig. 5. Deformation textures of the OFHC copper with Goss orientation at the strain of 70% under tension, compression, and simple shear.

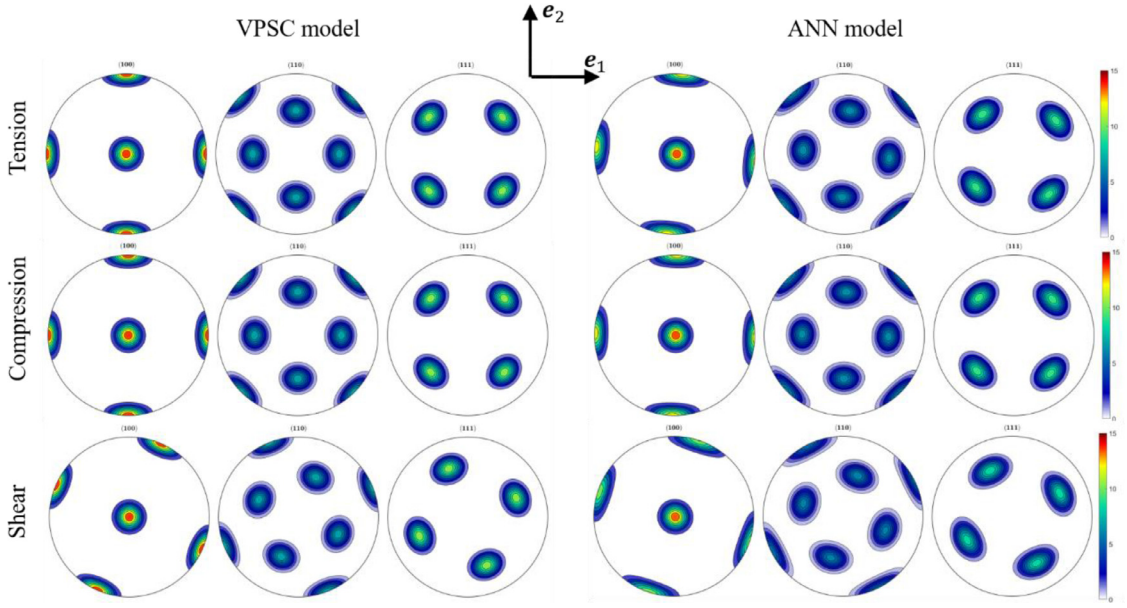


Fig. 6. Deformation textures of the OFHC copper with Cube orientation at the strain of 70% under tension, compression, and simple shear.

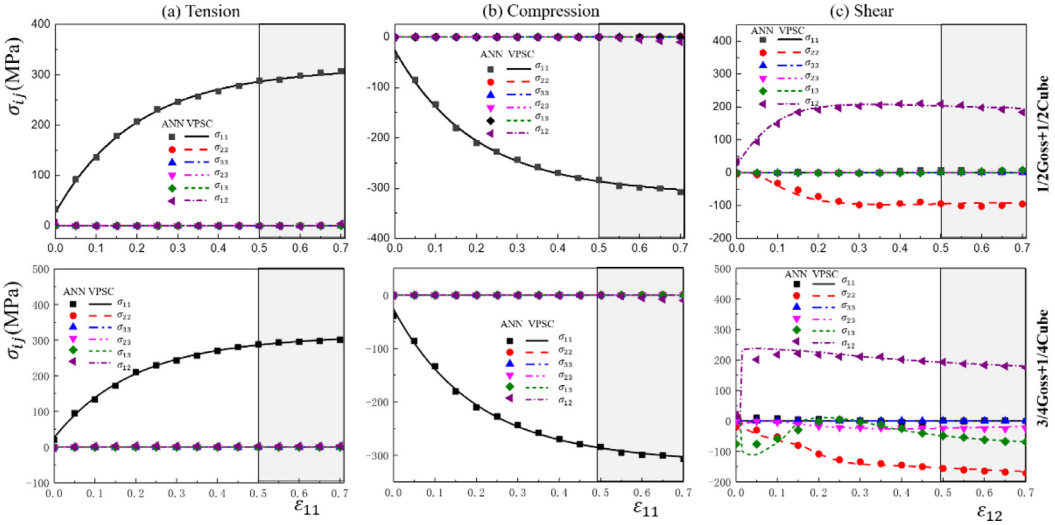


Fig. 7. Comparison between VPSC and ANN stress-strain curves under (a) uniaxial tension, (b) uniaxial compression, and (c) simple shear for 50% Goss+50% Cube (top row) and 75% Goss+25% Cube (bottom row) OFHC bicrystals.

for the ANN model. The training set is used to find the parameters to minimize the loss function between the output of the network and the target value, while the validation set is used to evaluate the performance of models with different hyperparameter values. The testing set is used for the assessment of the generalization error of the final chosen model. In addition, the testing set prevents the network from overfitting the training set. In the training stage, the mini-batch stochastic gradient descent (SGD) algorithm is employed as an optimizer with a learning rate of 0.001 and a momentum of 0.9. The networks are trained from scratch with a batch size of 64 for 500 epochs. The early stopping strategy is used to alleviate model overfitting and validation loss is monitored for stopping the model training. In the following sections, several examples are used to verify the generality of the trained model with respect to different loading conditions and initial textures.

3.1. Application to single crystals

ANN and VPSC simulations on the micromechanical behavior of the single-crystal OFHC copper with Goss (0°, 45°, 0°) and Cube (0°, 0°, 0°) orientations under three loading conditions up to 70% strain were conducted. It is also noted that the dataset generated for training and validating the ANN model only contains information within 50% strain. The predictions on the behavior of the OFHC copper beyond the 50% strain are used to benchmark the outreaching ability of the ANN model.

The stress components of Goss and Cube orientations under tension, compression, and simple shear predicted by the VPSC and ANN models are presented in Fig. 4 as a function of the primary strain. The ANN model reproduces well not only the evolution of the primary stress component but also that of the other non-primary stress components. The non-primary stress components σ_{22}, σ_{33}, σ_{12}, σ_{13}, and σ_{23} are all zero for tension and compression due to their prescribed boundary conditions (see Table 2). With respect to simple shear, the primary stress component σ_{12} and the non-primary stress component σ_{22} obtained from VPSC model are predicted by the ANN model. Moreover, the corresponding deformation textures predicted by VPSC and ANN models at the strain of 70% are compared in terms of the {100}, {110}, {111} pole figures in Figs. 5 and 6. The ANN model captures well the deformation

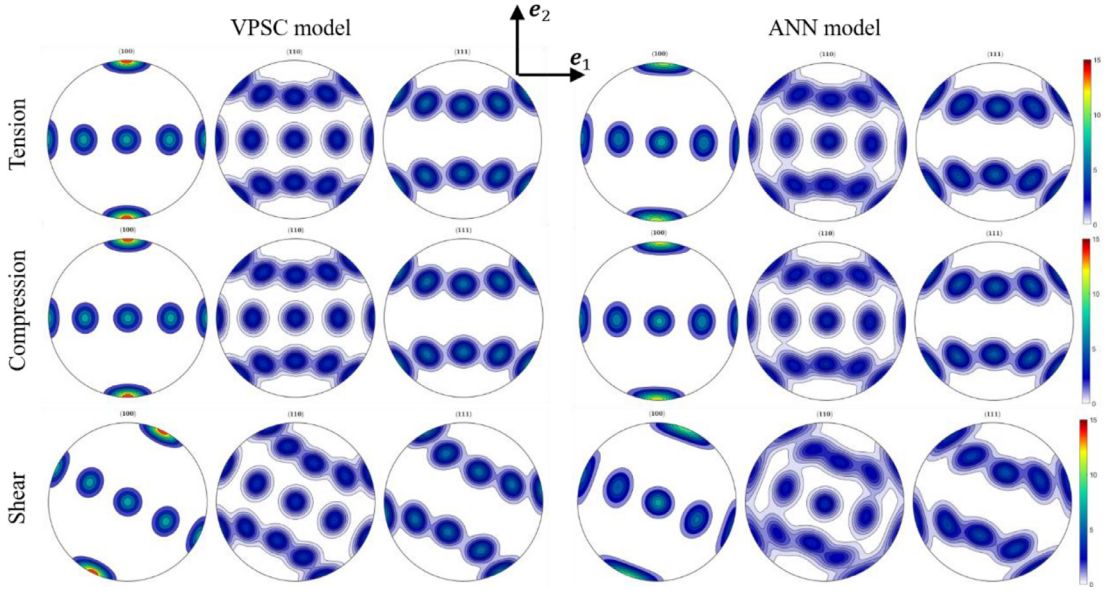


Fig. 8. Deformation textures of the OFHC copper bicrystal (50% Cube + 50% Goss) at the strain of 70% under tension, compression, and simple shear.

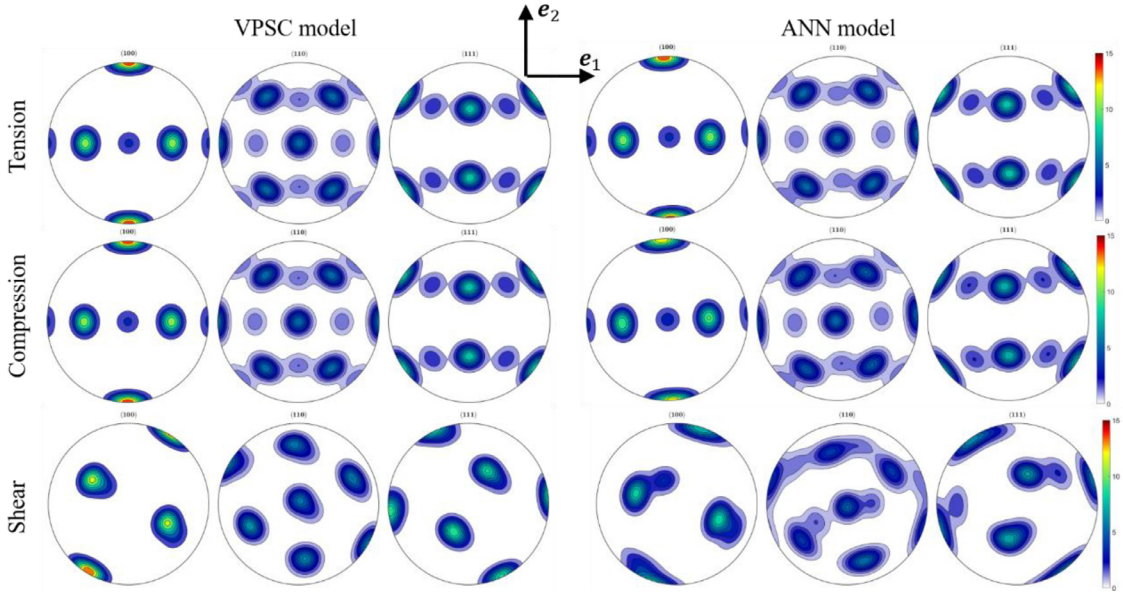


Fig. 9. Deformation textures of the OFHC copper bicrystal (75% Goss + 25% Cube) at the strain of 70% under tension, compression, and simple shear.

textures obtained in the VPSC model under various loading conditions. Especially for simple shear, where both the stress-strain response and the texture development of the two orientations are very different from each other. This grain orientation dependence is well captured by the trained ANN model. Relatively, it is much difficult for the ANN model to handle single grain because 96 grain orientations are required to represent the single grain, while the VPSC model merely uses three Euler angles. So the largest deviation between two models is foreseeable to be the cases of single crystal. However, it is encouraging that only few ANN pole figures deviates from the corresponding VPSC results by only very small and acceptable rotation (Fig. 5).

3.2. Application to bicrystals

The micromechanical behavior of two bicrystals, 50% Goss + 50% Cube and 75% Goss + 25% Cube, under various loading conditions, are simulated by the trained ANN model. Compared to the corresponding results by VPSC model, the ANN model reproduces well the evolution of all the stress components for both bicrystals (Fig. 7). For simple shear, the non-zero stress components are σ_{12} and σ_{22} for the bicrystal 50% Goss + 50% Cube and σ_{12}, σ_{22}, σ_{13} and σ_{23} for the bicrystal 75% Goss + 25% Cube. The distinct micromechanical behavior of the bicrystals under simple shear is ascribed to the anisotropy of the OFHC copper. In addition, the deformation textures predicted by VPSC and ANN models at the strain of 70% are compared in Figs. 8 and 9. Again, the ANN model reproduces well the deformation textures under various loading conditions.

3.3. Application to polycrystals

Weakly and strongly textured OFHC copper polycrystals, which respectively have random and rolled initial textures (Fig. 10), are considered. Tension, compression, and simple shear of these two polycrystals are modeled by the ANN model. Fig. 11 compares the

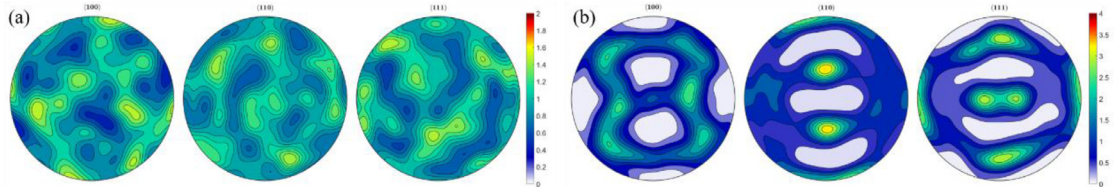


Fig. 10. Initial (a) random and (b) rolled textures in terms of {100, 110} and {111} pole figures.

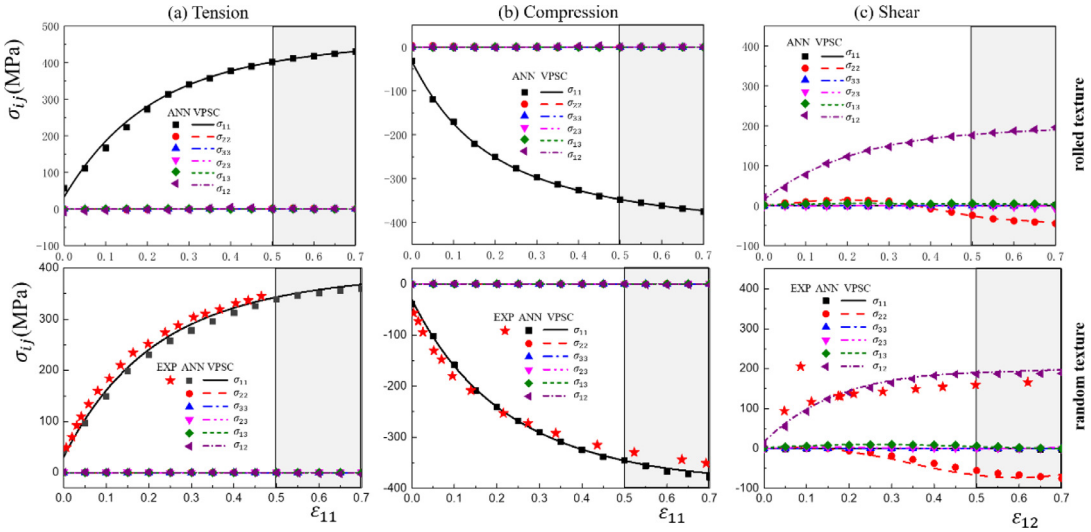


Fig. 11. Comparison between VPSC and ANN model stress-strain curves under uniaxial tension, uniaxial compression, and simple shear for rolled texture (top row) and random texture (bottom row).

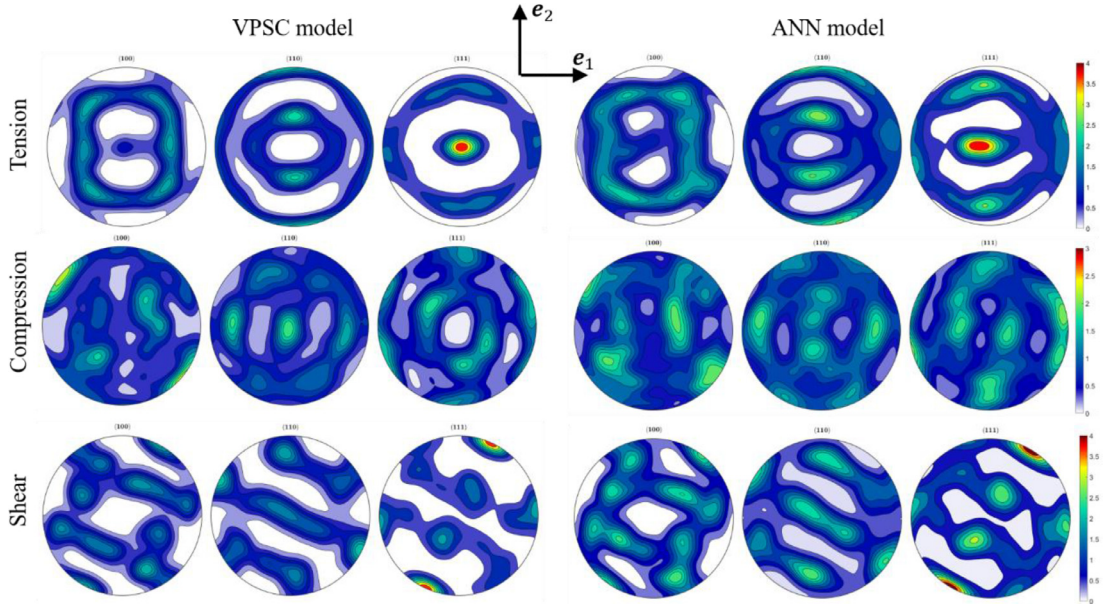


Fig. 12. Deformation textures of the OFHC copper at the strain of 70% for rolled texture under tension, compression, and simple shear.

ANN and VPSC stress-strain curves for the two polycrystals under three load conditions. The ANN model reproduces well not only the modeling results from the VPSC model, but also the available experimental results for the randomly textured polycrystal (asterisks in Fig. 11). The deformation textures of the rolled copper (not used in the training set) predicted by the ANN model are in good agreement with those obtained with the VPSC model (Fig. 12). In addition, the deformation textures of the randomly textured polycrystal under various loading conditions predicted by the ANN model are compared to the available experimental ones [26] (Fig. 13). The deformation textures at the tensile strain of 0.37, the compressive strain of -0.49, and simple shear strain of 0.43 are well reproduced by the ANN model.

Based on the above results, the trained ANN model reproduces well all the results from the VPSC model in terms of micromechanical behavior and texture evolution and their dependency on the initial texture and loading conditions. As the reliable VPSC model is based on, the trained ANN model captures well the experimen-

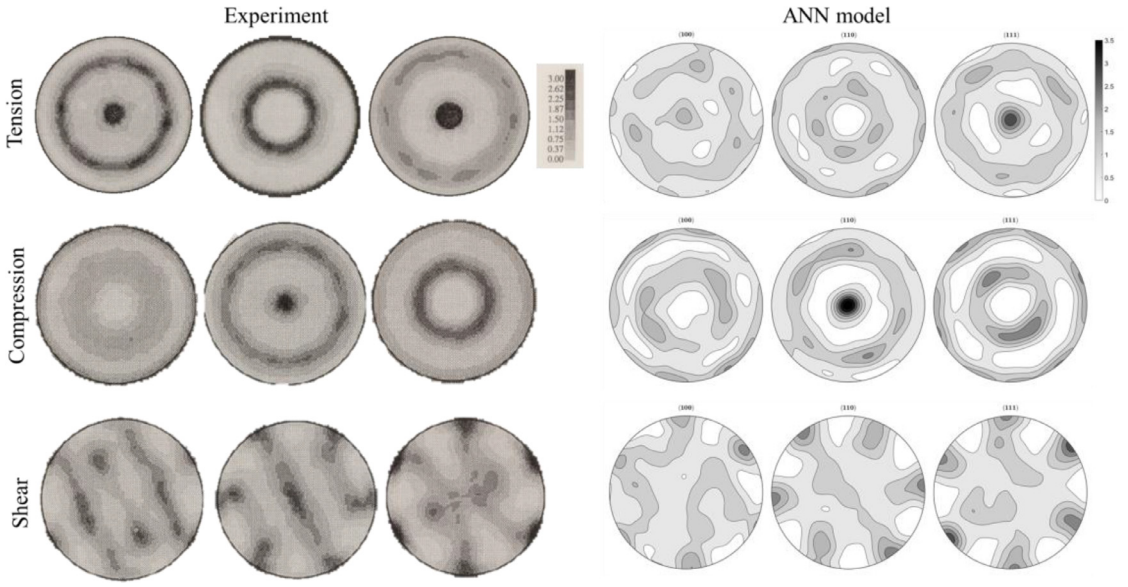


Fig. 13. Experimental and ANN predicted deformation textures (in terms of the {100}, {110, 111} pole figures) of the OFHC copper at the tensile strain of 0.37, the compressive strain of -0.49, and the simple shear of 0.43. The experimental pole figures are taken from Bronkhorst et al. [26].

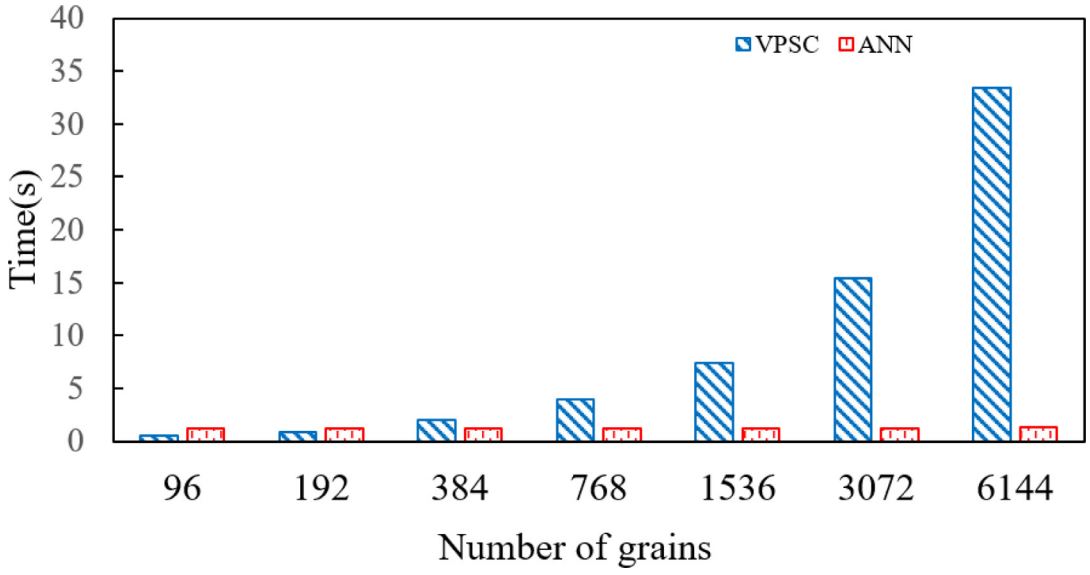


Fig. 14. Runtime comparison between VPSC and the proposed ANN model.

tally investigated micro-mechanical behavior of the OHFC copper under various loading conditions.

3.4. Computational efficiency of the ANN model

It is well known that the physics-based crystal plasticity simulations demand a high computational cost for the highly non-linear differential equations [39]. Compared to the full-field polycrystal models (e.g., CPFEM, [1,40,11]), the VPSC model, a homogenized polycrystal model used in the current work, is already computationally much more efficient. However, the VPSC model is still not efficient enough for broad engineering applications.

It is also well known that the machine learning-based approach has the potential to significantly improve computational efficiency. Once an ANN model is trained and validated, it does not require the expensive crystal plasticity simulations anymore. Therefore, crystal plasticity-based VPSC simulations and the proposed ANN model were compared to analyze the computational advantage of using ANN models.

Fig. 14 compares the runtimes between the VPSC and ANN simulations with different grains under uniaxial tension. The runtime of the VPSC model increases dramatically with increasing the grain number, while that of the ANN model is nearly independent of the grain number. With increasing the grain number, the saving of the computational cost can be significantly enhanced. It is envisioned that the proposed ANN model, together with a conventional finite element approach, can efficiently model the micromechanical behavior of a material with complex macro- and micro- structures under complex loading conditions.

5. Conclusion

A machine learning-based ANN model is proposed for the micromechanical behavior of OFHC copper under various loading conditions up to very large deformation. As demonstrated, the ANN

model is capable of capturing the highly non-linear response of the single crystals, bi-crystals, and polycrystals of OFHC copper. The ANN model not only reproduces the results produced by the VPSC model but also those measured experimentally. These results include the stress-strain relation and texture evolution and their dependency on the initial texture and loading conditions. The success of this approach motivates further use of the machine learning-based technique as a powerful tool in the advancement in both the academic fields (e.g., materials science, mechanical engineering) and industrial applications. To name a few possible ANN model-based numerical experiments on various metallic materials: mechanical tests, typical processing techniques of rolling, extrusion, folding, etc., and fabrication technologies with severe plastic deformation of equal channel angular extrusion, torsional extrusion, etc.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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